

Course: B. Tech.

Branch: Computer Engineering

Semester: IV

Subject Code & Name: Numerical Methods [BTCOE406B_Y19]

Max Marks: 60

Date: 29-01-24

Duration: 03:00 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/ Marks
CO)

Q. 1 Attempt the following questions.

[12]

- A)** Find the root of the equation $x^3 - 4x - 9 = 0$ using the bisection method correct to three decimal places. **CO1**
- B)** Find the positive root of $x^4 - x = 10$ correct to three decimal places using Newton-Raphson method **CO1**

Q.2 Solve any TWO Questions.

[12]

- A)** Apply Gauss elimination method to solve the equations $x + 4y - z = -5$, $x + y - 6z = -12$, $3x - y - z = 4$ (Solve using simultaneous equation method). **CO2**
- B)** Apply Gauss-Jordan method to solve the equations (Use Matrix method) **CO2**
 $x + y + z = 9$, $2x - 3y + 4z = 13$, $3x + 4y + 5z = 40$.
- C)** Solve by Jacobi's iteration method, the equations $10x + y - z = 11.19$, $x + 10y + z = 28.08$, $-x + y + 10z = 35.61$, correct to two decimal places. **CO2**

Q. 3 Solve any TWO Questions.

[12]

- A)** Evaluate the following: **CO3**
 i) $\Delta \tan^{-1}x$ ii) $\Delta (e^x \log 2x)$ iii) $\Delta^2 \cos 2x$
- B)** Find the missing term in the table: **CO3**

x	:	2	3	4	5	6
y	:	45.0	49.2	54.1		67.4

- C)** In the table below, the values of y are consecutive terms of a series of which 23.6 is the 6th term. Find the first and tenth term of the series: **CO3**

x	:	3	4	5	6	7	8	9
y	:	4.8	8.4	14.5	23.6	36.2	52.8	73.9

Q.4 Attempt the following questions.

[12]

A) Derive the Newton-Cotes Quadrature formula for the area cover between $\int_a^b f(x) dx$. **CO4**

B) Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using **CO4**

i) Trapezoidal Rule ii) Simpson's 1/3 rule iii) Simpson's 3/8 rule

Q. 5 Attempt the following questions.

[12]

A) Find the value of y for x = 0.1 by Picard's method, given that **CO5**

$$\frac{dy}{dx} = \frac{y-x}{y+x} \quad y(0)=1$$

B) Apply Runge-Kutta fourth order method, to find an approximate value of y when x = 0.2, given that $dy/dx = x + y$ and $y = 1$ when $x = 0$. **CO5**

******* END OF PAPER *******